

Assembly guide — ROSOTA leader gripper (2-DOF wrist + grasp levers)

0. Before you start

Print every part in [STL/printable_right/](#) for a right-hand unit, or [STL/printable_left/](#) for a left-hand unit. A full console needs one of each.

setting	value
material	PLA (Bambu PLA Basic)
layer height	0.20 mm
walls / infill	2 walls, 15 % cubic infill
supports	on

Fig. 1 shows the assembled right-hand gripper with every printed part named.

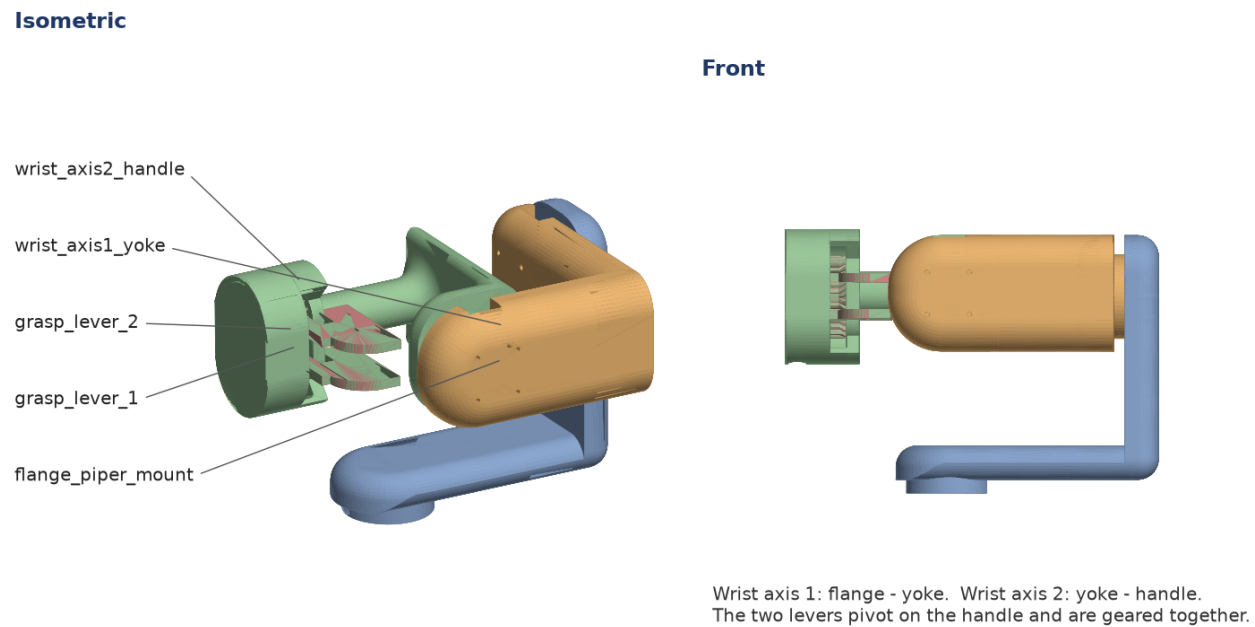


Fig. 1. Assembled right-hand gripper (meshes from STL/urdf_meshes placed at the URDF link origins).

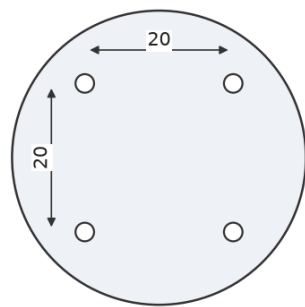
1. Flange to the arm

- 1 Bolt `flange_piper_mount` onto the PiPER `link6` face through the four holes on PCD 20 mm in its base plate (Fig. 3).
- 2 Check that the flange seats flat — this joint is fixed in the URDF, so any tilt here shows up as a constant offset in every wrist reading.

2. Wrist axis 1

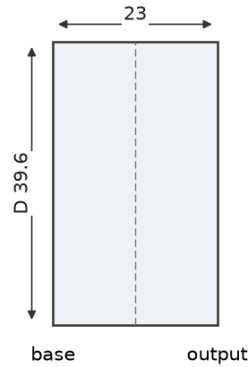
The MF4005-V2 mounting faces are shown in Fig. 2.

Motor base (fixed side)

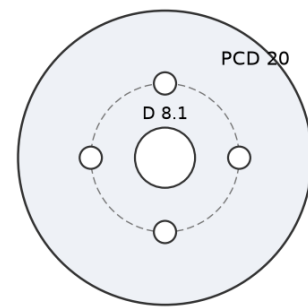


4 x M2.5 on a 20 x 20 mm square
bolts to wrist_axis1_yoke

Side



Motor output (rotating side)



4 x M3 on PCD 20 mm
servo 1 -> flange_piper_mount
servo 2 -> wrist_axis2_handle

MF4005-V2 mounting faces, from the vendor installation drawing. Dimensions in mm.

Fig. 2. MF4005-V2 mounting faces and the printed part each face is fixed to.

- 1 Seat the first **MF4005-V2** servo in `wrist_axis1_yoke`, motor base against the yoke.
- 2 Attach the servo output to `flange_piper_mount`.
- 3 Set the motor's CAN ID **before** it goes into the bundle — the right wrist uses IDs **2 and 4**, the left wrist IDs **2 and 6**. MF4005 replies on the same arbitration ID it is queried on (`0x140+ID`), so two motors sharing an ID on one bus cannot be told apart afterwards.

3. Wrist axis 2

- 1 Seat the second MF4005-V2 in `wrist_axis1_yoke`.
- 2 Attach `wrist_axis2_handle` to its output.
- 3 Route both motor cables along the outside of the handle.

4. Grasp module

- 1 Fit `grasp_lever_1` and `grasp_lever_2` onto `grasp_mount_plate` with `grasp_pin_1` / `grasp_pin_2`.
- 2 Mesh the lever gear teeth so the two levers move symmetrically. The console software reads the pair as a single grasp angle, so a half-tooth offset is a permanent bias.
- 3 Mount the magnetic rotary encoder on the lever shaft. Set the magnet gap so the reading changes smoothly across the full lever travel — a reading that never moves by more than one LSB means the magnet is too far away or not centred.
- 4 Fit `grasp_cover`.

5. Electronics

- 1 Mount the **Seeed XIAO nRF52840 Sense** on `mcu_base`.
- 2 Wire the encoder to the XIAO.
- 3 The XIAO enumerates as USB `2886:8045` and streams the grasp angle in degrees over USB CDC.

6. Checks before use

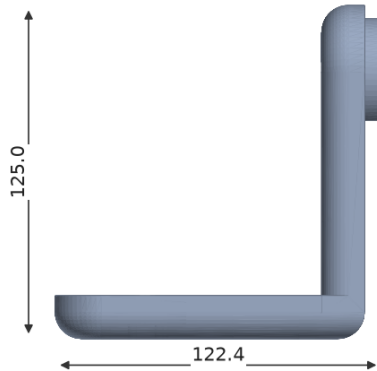
- [] Both MF4005 answer on their intended IDs, on the intended bus.
- [] Bus voltage reads 12.3–12.5 V under load.
- [] The encoder sweeps its full range over the lever travel, with no jumps.
- [] Assembled gripper mass is close to 207 g — a large difference means a part was printed with different settings, and the URDF inertials will be off.

7. Reference drawings

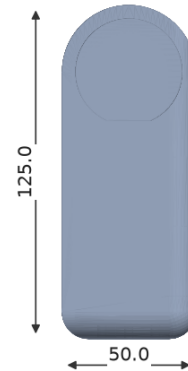
Overall dimensions of the right-hand parts. The left-hand parts in [STL/printable_left/](#) are their mirror images.

Flange - PiPER link6 mount — flange_piper_mount

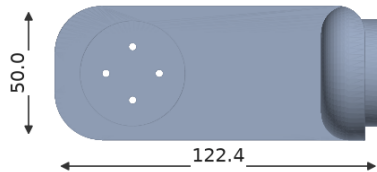
Front



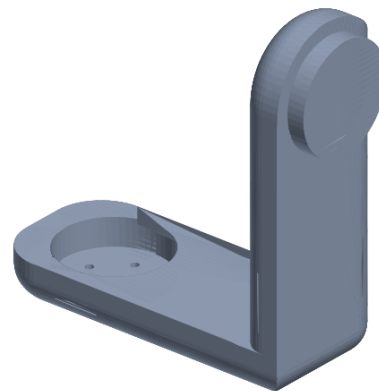
Right side



Top



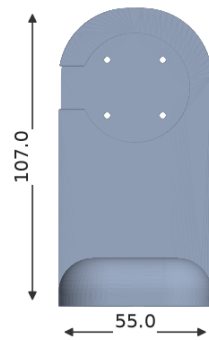
Isometric



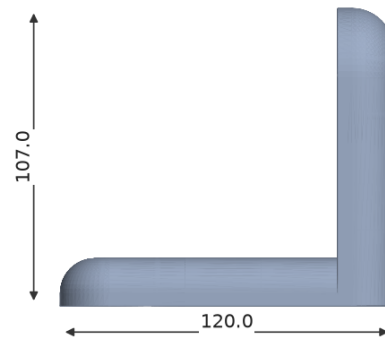
PiPER end: 4 x D3.0 holes on PCD 20 mm in the base plate (link6 side). Servo 1 sits on the round pad at the top of the upright.
Dimensions in mm, overall extents of the right-hand part.

Fig. 3. flange_piper_mount.

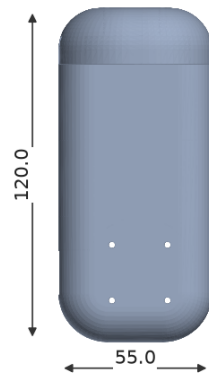
Wrist axis-1 yoke — wrist_axis1_yoke **Front**



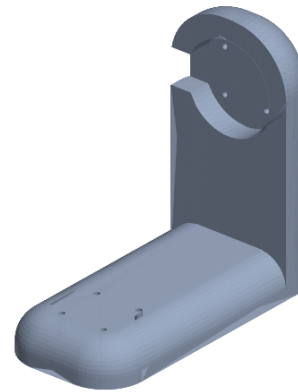
Right side



Top



Isometric

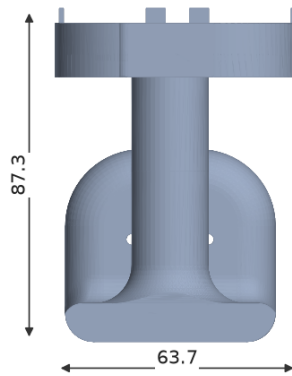


Bases of both servos: 2 x (4 x M2.5 on 20 x 20 mm), on perpendicular faces.
 Dimensions in mm, overall extents of the right-hand part.

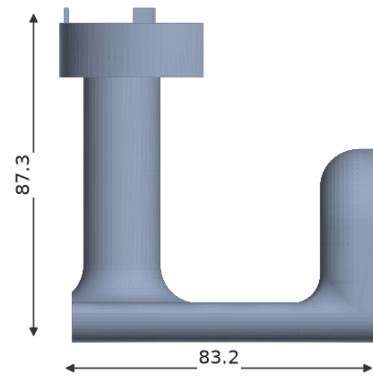
Fig. 4. wrist_axis1_yoke.

Wrist axis-2 handle — wrist_axis2_handle

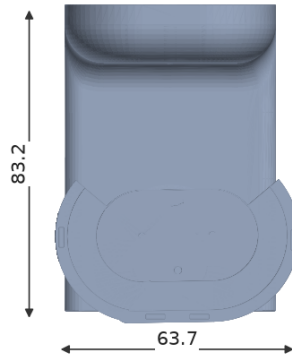
Front



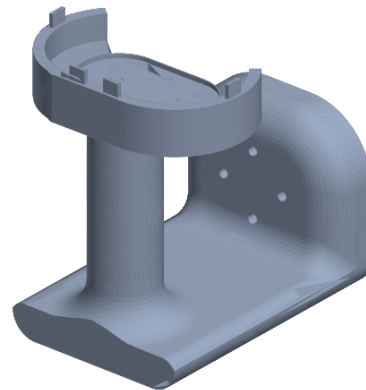
Right side



Top



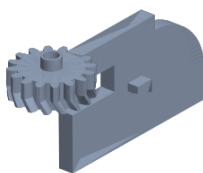
Isometric



Output face of servo 2: 4 x M3 on PCD 20 mm.
Dimensions in mm, overall extents of the right-hand part.

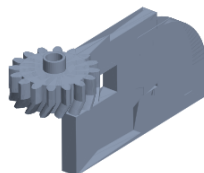
Fig. 5. wrist_axis2_handle.

Grasp lever 1



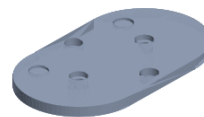
grasp_lever_1
20.5 x 62.3 x 32.0 mm

Grasp lever 2



grasp_lever_2
20.5 x 62.7 x 32.0 mm

Grasp mount plate



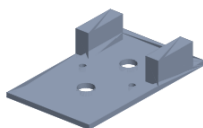
grasp_mount_plate
24.0 x 42.5 x 5.0 mm

Grasp pin 1 / 2



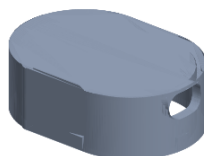
grasp_pin_1
2.0 x 2.1 x 10.5 mm

Grasp sensor base



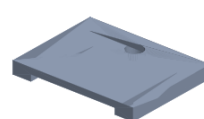
grasp_sensor_base
30.0 x 50.0 x 9.5 mm

Grasp cover



grasp_cover
63.7 x 45.6 x 21.0 mm

MCU base



mcu_base
30.0 x 23.0 x 3.0 mm

Fig. 6. Grasp-module parts.